P P SAVANI UNIVERSITY

Seventh Semester of BCA & B.Sc.IT Examination

October 2026

SSCA4010 Augmented Reality & Virtual Reality

13.10.2026, Tuesday **Time: 12:30 p.m. To 3:00 p.m.** **Maximum Marks: 60**

Instructions:

1. The question paper comprises two sections.
2. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

			CO	BTL
Q - 1	MCQ/Short Question/Fill in the Blanks (Any Five)	**[05]**		
(i)	XR stands for _____ and is the umbrella term for AR, VR and MR.		**1**	**1**
(ii)	Which type of AR uses a predefined image such as a poster or QR code to trigger digital content? (a) Marker-less AR (b) Marker-based AR (c) Projection Mapping (d) Virtual Reality		**1**	**1**
(iii)	A device that tracks only pitch, yaw and roll provides _____ degrees of freedom.		**1**	**1**
(iv)	Which technology fully replaces the user's physical surroundings with a computer-generated environment? (a) Augmented Reality (b) Mixed Reality (c) Virtual Reality (d) Projection Mapping		**1**	**1**
(v)	List the three rotational movements tracked in 3DOF.		**1**	**1**
(vi)	Lens Studio is a free AR development platform developed by _____.		**2**	**1**
(vii)	To publish a Lens, its total file size must be under: (a) 2 MB (b) 8 MB (c) 6 MB (d) 24 MB		**2**	**1**
Q - 2	Attempt any two.	**[10]**		
(i)	Define Augmented Reality. Explain the three types of AR: Marker-based, Marker-less and Projection Mapping with one real-world use case for each.		**1**	**2**
(ii)	Differentiate between AR and VR on the basis of immersion, hardware required, spatial awareness, computing power and application areas. Give one example application of each.		**1**	**4**
(iii)	What is Degrees of Freedom (DOF)? Explain 3DOF and 6DOF tracking. Which one is better suited for an AR furniture-placement app? Justify.		**1**	**3**
(iv)	Write a short note on Extended Reality (XR). Explain how AR, VR and MR fit under XR with one industry example of each.		**1**	**2**
Q - 3	Attempt any two.	**[10]**		
(i)	What is Lens Studio? Where do users add assets, what does the preview panel do, and what is the Scene Hierarchy? What is the use of the Inspector Panel?		**2**	**2**
(ii)	Explain the steps to publish a Lens from Lens Studio. List the mandatory requirements for submission such as Lens name, Lens icon, preview video and size limits.		**2**	**2**
(iii)	What is the Asset Library in Lens Studio? Explain how it helps a Lens creator, with examples of the types of assets that can be added from it.		**2**	**2**
Q - 4	Attempt any one.	**[05]**		
(i)	An eyewear brand wants customers to scan its product catalogue page so		**3**	**3**

that a 3D model of the sunglasses appears on top of it. Explain how Marker Tracking in Lens Studio can be used to build this, what makes a good marker image, and describe the user experience.

(ii)	A museum wants a life-size 3D dinosaur to appear on the floor of the exhibit hall and walk around visitors when they point their phone at it. Explain how World Tracking (World Mesh) in Lens Studio can be used to build this, how it differs from marker-based AR, and describe the user experience.	3	3	

SECTION – II

Q - 1	MCQ/Short Question/Fill in the Blanks (Any Five)	[05]	CO	BTL
(i)	Which camera in Lens Studio keeps parallel lines parallel regardless of their distance from the camera? (a) Perspective Camera (b) Orthographic Camera (c) Fisheye Camera (d) Depth Camera		3	1
(ii)	The _____ Camera is commonly used for 2D effects such as Screen Image and Screen Text.		3	1
(iii)	Which tracking type allows a Lens to place a 3D object on the floor or a table around the user? (a) Face Tracking (b) Hand Tracking (c) World Tracking (d) Body Tracking		3	1
(iv)	Bitmoji is a personalised cartoon _____ of a Snapchat user.		4	1
(v)	Which type of tracking would you use to place a virtual hat on a user's head?		3	2
(vi)	Which Lens Studio script lets you create simple trigger–response interactions (e.g. tap to play a sound) without writing code? (a) Tween Script (b) Behavior Script (c) Shader Script (d) Physics Script		4	1
(vii)	In the _____ Camera, objects appear smaller as they move further away.		3	1
Q - 2	Attempt any two.	[10]		
(i)	Explain the following types of tracking - Face, Hand, Body and World Tracking supported by Lens Studio with one Lens idea for each.		3	2
(ii)	Explain how Marker Tracking works in Lens Studio. What makes a good marker image? Give one use case such as a product package or event poster.		3	2
(iii)	Differentiate between the Perspective Camera and the Orthographic Camera in Lens Studio. When would you use each? Give examples.		3	4
(iv)	Explain how to add an Image, Text and Audio to a Lens Studio project. Give an example of a Lens that uses all three.		3	3
Q - 3	Attempt any two.	[10]		
(i)	Write a short note on Lens Studio's Generative AI Suite. Explain 3D Asset Generation and Character Animation with one use case for each.		4	2
(ii)	What is Bitmoji? Explain the Bitmoji features available in Lens Studio and describe one Lens experience that uses the user's Bitmoji avatar.		4	2
(iii)	What is the Behavior Script in Lens Studio? Explain its trigger and response system with suitable examples, e.g. tapping the screen to play an audio or to show/hide a 3D object.		4	3
Q - 4	Attempt any one.	[05]		
(i)	Design a Lens concept for a college fest: when a student scans the fest		4	6

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poster, an animated 3D mascot appears, plays a welcome audio and shows event details as text. Explain the tracking, camera, assets and scripts you would use.

(ii)	A jewellery brand wants customers to virtually try on a ring and a wristwatch using the phone camera. Explain how Hand Tracking in Lens Studio can attach these 3D objects to specific hand joints (such as the wrist and finger joints), and describe the user experience.	**3**	**3**

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CO : Course Outcome Number BTL : Blooms Taxonomy Level

Level of Bloom's Revised Taxonomy in Assessment

1: Remember	2: Understand	3: Apply
4: Analyze	5: Evaluate	6: Create

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